

# Announcements

- Maven Activity and Project 1 Progress Report is Due this Sunday
- You should now start working on the Project 1 Requirements Assignment
- REST assignment was out last Thursday - This is an individual assignment. Due Next Sunday.
- Extra Credit Opportunity in today's class

CS3300 Introduction to Software Engineering

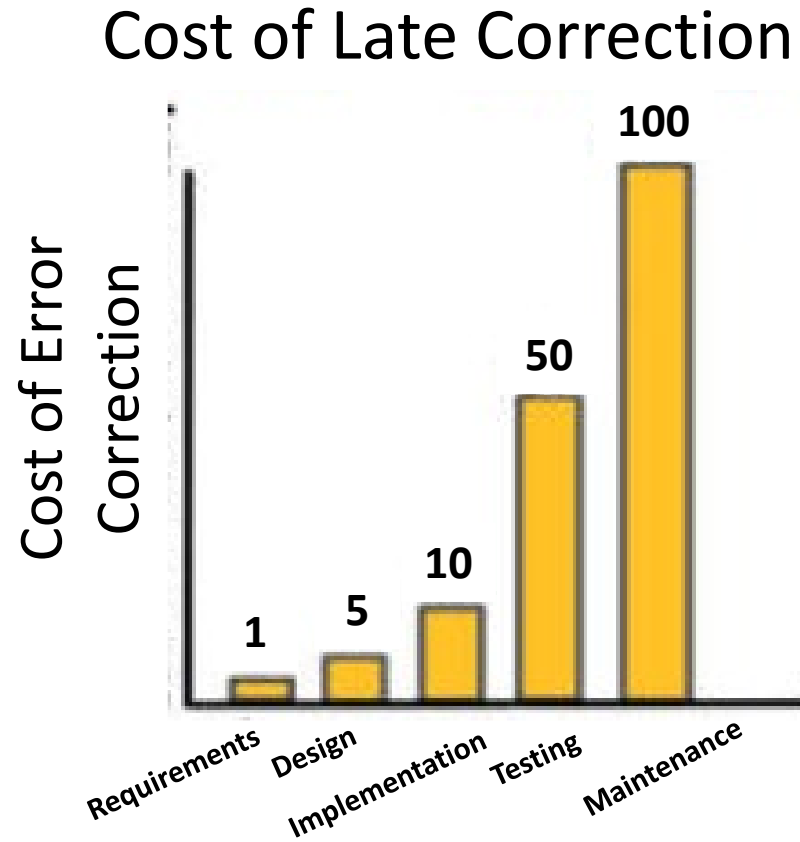
# Lecture 07: Requirements Engineering

Dr. Nimisha Roy ▶ [nroy9@gatech.edu](mailto:nroy9@gatech.edu)

# Requirements Engineering (RE)

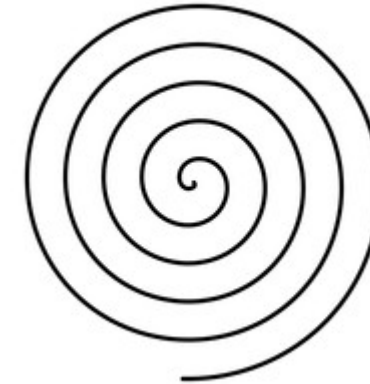


RE is the process of establishing the needs of stakeholders that are to be solved by software



Management

Elicitation

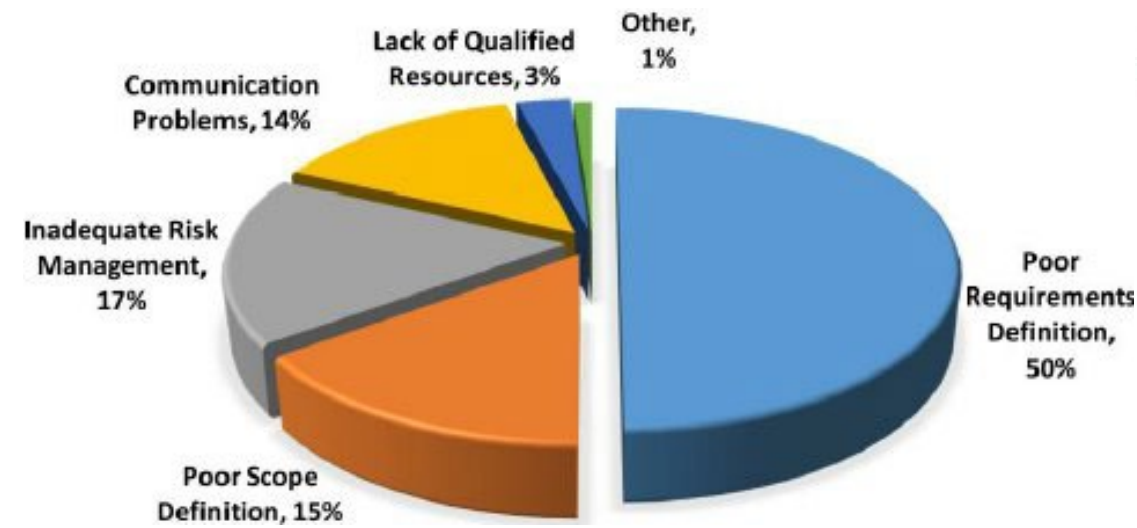


Analysis

Validation

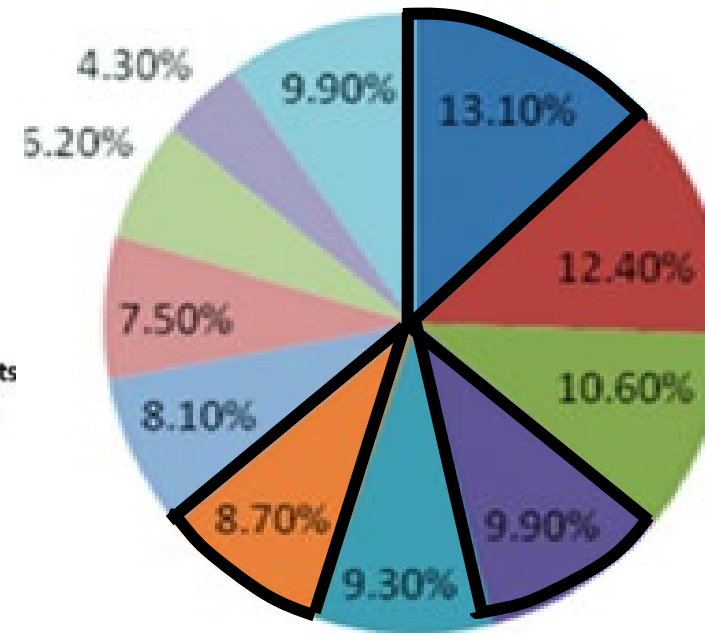
Specification

# Software Failures due to RE



## Role of Requirements in Software Project Failures (Agile)

Source: Abdou, T., Kamthan, P., & Shahmir, N. (2014). User Stories for Agile Business: INVEST, Carefully!. Social Media and Publicity, 141.



## Role of Requirements in Software Project Failures.

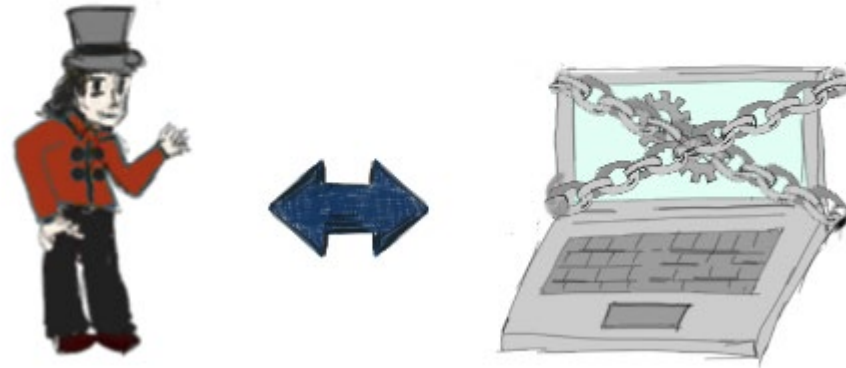
Source: Hussain, A., Mkpojiogu, E. O., & Kamal, F. M. (2016). The role of requirements in the success or failure of software projects. International Review of Management and Marketing, 6(7S), 306-311

- Incomplete Requirements
- Lack of user involvement
- Lack of Resources
- Unrealistic Expectations
- Lack of Executive Support
- Changing Requirements
- Lack of planning
- Not needed any longer
- Lack of IT Management
- Technology Illiteracy
- Others

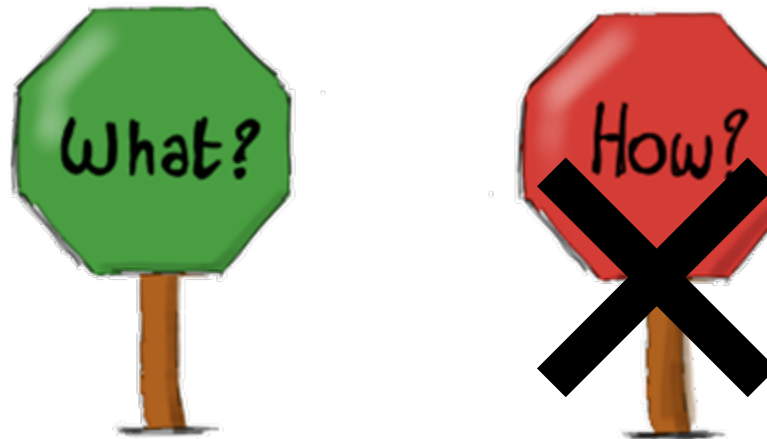
# Requirements Engineering (RE)



= =

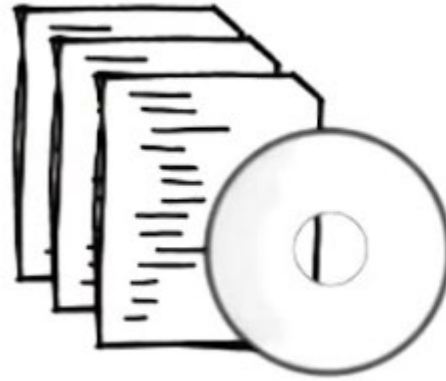


=> Software Requirements Specification (SRS)



# Software Intensive Systems

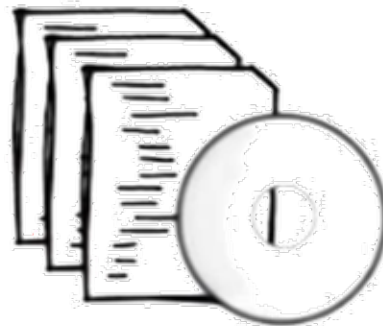
Software



Software Intensive System = Software + Hardware + Context



=



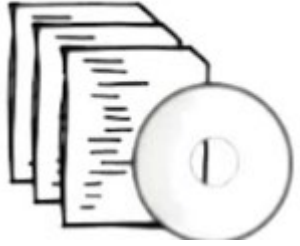
+



+



# Software Quality



Software runs on some hardware and is developed for a purpose that is related to human activities

$$\text{Quality} \neq f(\text{stack of papers and CD})$$

$$\text{Quality} = f(\text{stack of papers and CD}, \text{target with dart})$$



RE is mostly about identifying the purpose

# Identifying Purpose = Defining Requirements



Extremely Hard Task

- Sheer Complexity of the purpose/requirements
- “Often, people don’t know what they want until you show it to them” – Steve Jobs
- Changing requirements
- Multiple stakeholders with conflicting requirements

# Software Quality & RE: ICU Monitoring System

**System:** Real-time patient monitoring system that tracks vital signs and alerts healthcare professionals when readings become abnormal.

**Quality depends on its context:**

- **Software:** Process real-time data accurately and generate appropriate alerts.
- **Hardware:** Reliable sensors and monitors must operate continuously.
- **Context:** ICU is a high-stakes, 24/7 environment where delayed or inaccurate alerts can have serious consequences.

# Software Quality & RE: ICU Monitoring System

Requirements Engineering translates that context into requirements:

- **Functional:** The system shall alert healthcare professionals when a vital sign exceeds its defined safe range.
- **Non-functional:** Alerts shall be generated within **1 second** of detecting an abnormal reading.
- **Reliability:** Downtime shall be less than **0.1%**.
- **Integration/constraints:** Must integrate with existing hospital systems and satisfy applicable privacy requirements.

Takeaway: Quality = fitness for purpose → understanding the context helps us define the right requirements.

# Definition of Requirements Engineering

Requirements Engineering (RE) is a set of activities concerned with identifying and communicating the purpose of software-intensive system, and the context in which it will be used. Hence, RE acts as the bridge between the real-world needs of users, customers, and other constituencies affected by a software system, and the capabilities and opportunities afforded by software – intensive technologies.

Communication is as important as analysis

Quality means fitness-for-purpose. Cannot say anything about quality unless you understand the purpose

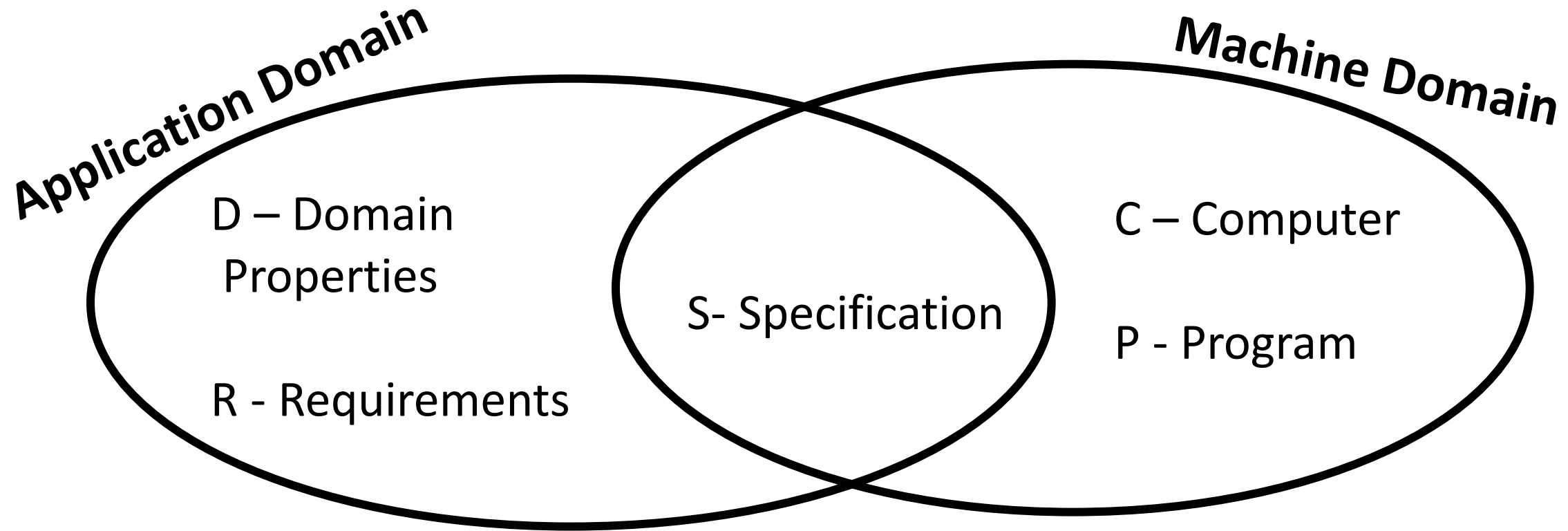
Needed to identify all stakeholders – not just the customer or user

Designers need to know how and where the system will be used

Requirements are partly about what is needed...

... and partly about what is possible

# RE is a shared phenomenon between domains



- Machine Domain - hardware/OS/libraries
- Application Domain - world in which software will operate
- Events in real world that machine can detect – buttons pushed
- Actions in real world that machine can cause- image appearing on screen



# Requirements Elicitation: Where do Requirements come from?

Stakeholders



Application Domain



Documentation



# Requirements Elicitation: Other Techniques

## Collaborative Techniques - Brainstorming

A software company is developing a new mobile game and invites both developers and gamers for a brainstorming session. During the session, a suggestion for a unique game mode emerges that becomes a central feature of the final product.



## Social Approaches - Ethnographic techniques

Analysts spend a day in a busy restaurant to observe how servers use the point-of-sale system. They notice that servers often bypass a time-consuming screen, jotting orders down on paper to enter later, indicating a need for a more efficient user interface.



# Requirements Elicitation: Other Techniques

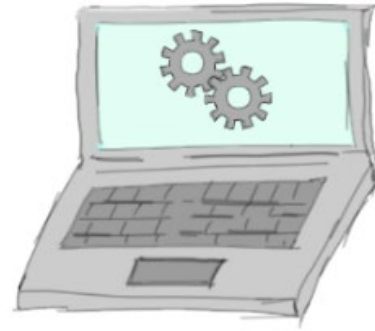
**Cognitive techniques** - Problem solving methods,  
Prototyping, Cognitive Task Analysis

In designing software for air traffic controllers, analysts study the controllers' processes for tracking multiple planes. They find that certain visual cues and alerts can reduce mental strain

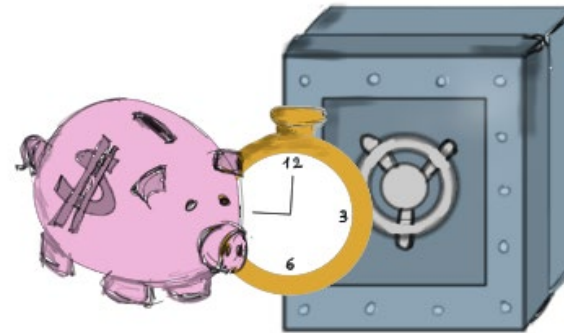


# Requirements Elicitation: Functional and Non-functional Requirements

Functional



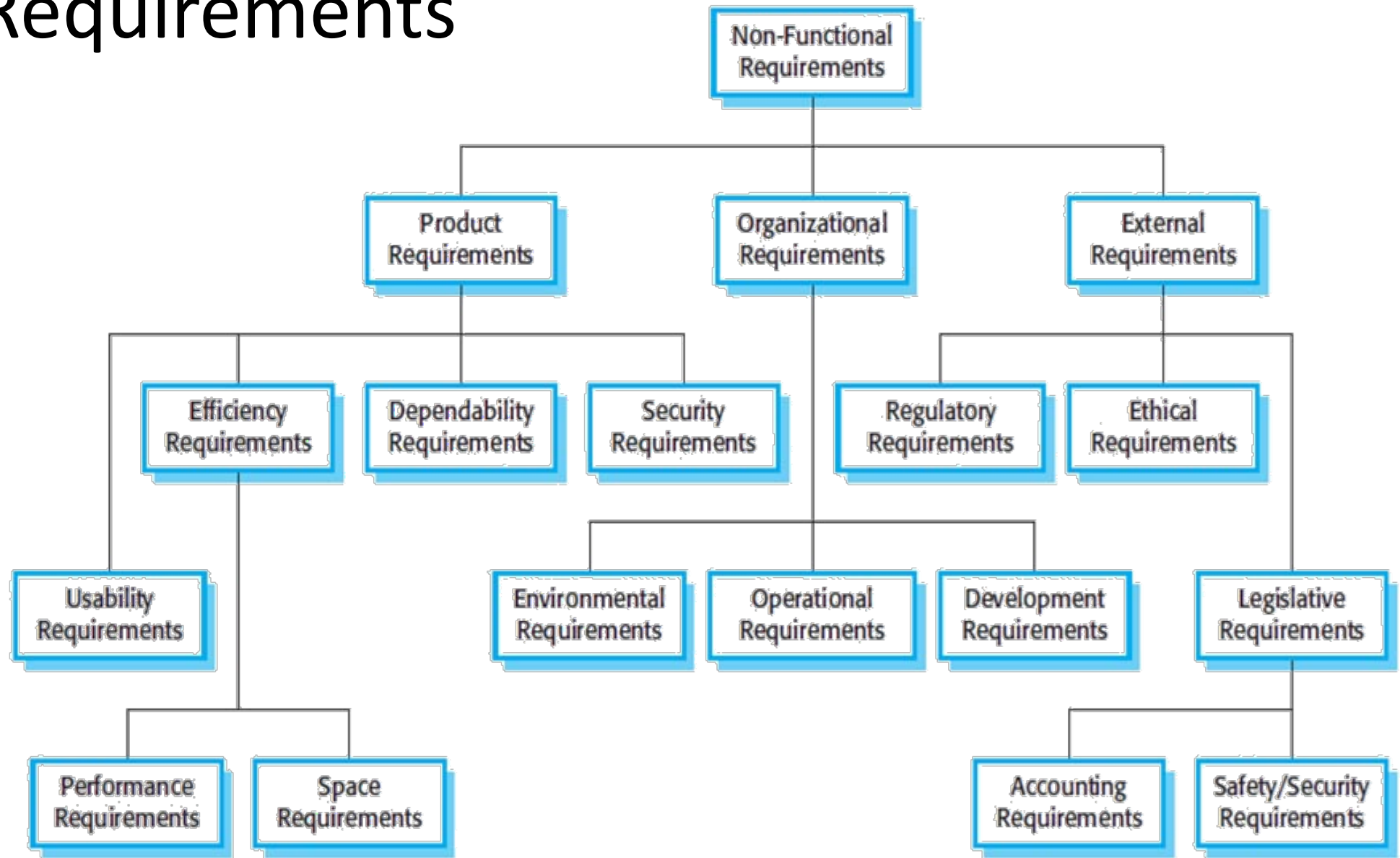
Non-Functional



**Non-Functional requirements:** criteria that can be used to judge the operation of a system, rather than specific behaviors. In essence, they describe how the system works, rather than what it does. E.g. security, accuracy, performance, cost, usability, adaptability, interoperability, reusability and so on.

# Requirements Elicitation: Functional and Non-functional Requirements

***Non-Functional Requirement may be more critical than functional requirement***



# Requirements should be measurable



| Property    | Measure                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| Speed       | Processed transactions/second<br>User/event response time<br>Screen refresh time                                   |
| Size        | Mbytes<br>Number of ROM chips                                                                                      |
| Ease of use | Training time<br>Number of help frames                                                                             |
| Reliability | Mean time to failure<br>Probability of unavailability<br>Rate of failure occurrence<br>Availability                |
| Robustness  | Time to restart after failure<br>Percentage of events causing failure<br>Probability of data corruption on failure |
| Portability | Percentage of target dependent statements<br>Number of target systems                                              |

# An example: MHC-PMS: A patient management system for mental health care

- This case study describes a real system (although that is not its real name) which was used (and may still be used) in a number of UK hospitals, including hospitals in Scotland.
- The system is designed for use in clinics attended by patients suffering from mental health problems and records details of their consultations and conditions.
- It is separate from a more general patient records system as more detailed information has to be maintained and the system has to be set up to generate letters and reports of different types and to help ensure that the laws pertaining to mental health are maintained by staff treating patients.

# MHC-PMS: Non functional Requirements

- ***Product requirement*** The MHC-PMS shall be available to all clinics during normal working hours (Mon–Fri, 0830–17.30). Downtime within normal working hours shall not exceed five seconds in any one day.
- ***Organizational requirement*** Users of the MHC-PMS system shall authenticate themselves using their health authority identity card.
- ***External requirement:*** The system shall implement patient privacy provisions as set out in the country's law.

# Example Quiz

A software application designed for a hospital is running smoothly on its intended hardware. However, the nurses using the system report that it takes too long to enter patient information, causing delays in patient care. Based on the definition of software quality discussed, what type of requirement might need improvement?

- A) Functional Requirement
- B) Non-Functional Requirement
- C) Maintenance Requirement
- D) Hardware Requirement

# Requirements Analysis: Traditional Techniques



Background Reading



Hard Data and Samples



Interviews

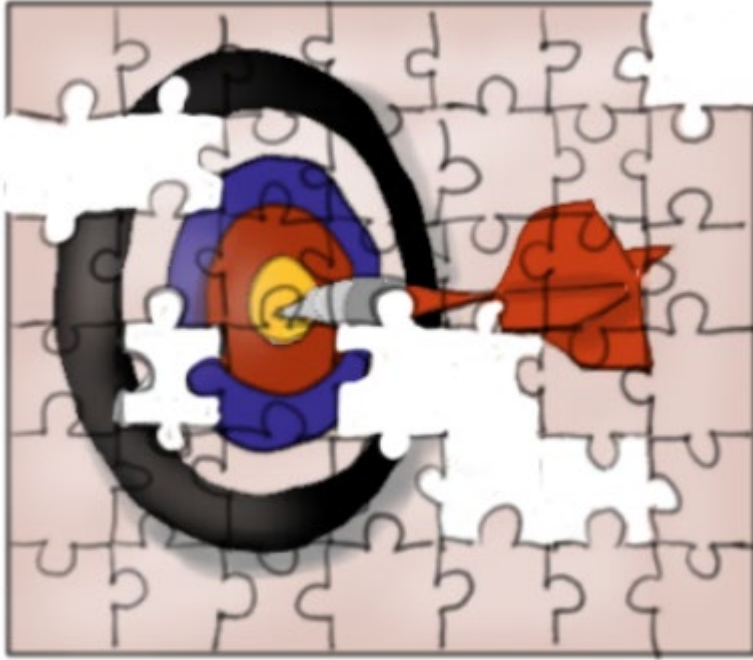


Surveys

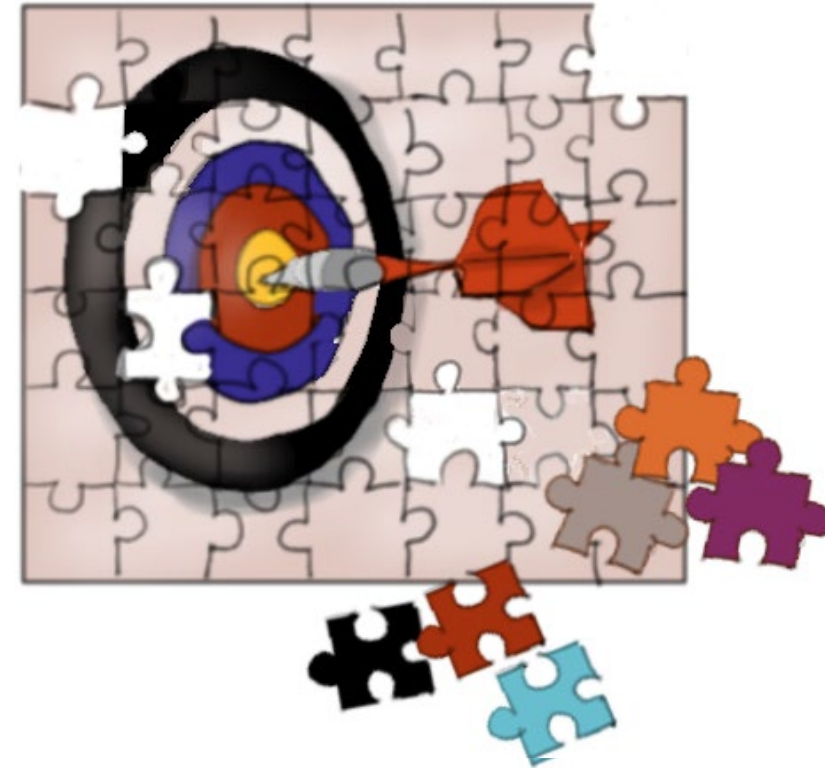


Meetings

# Requirements Analysis: Completeness and Pertinence

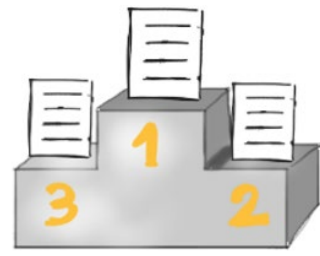


Difficult to identify all requirements, incomplete requirements collected and S/W missing functionality



- Relevance of requirements;
- Irrelevant conflicting requirements collected for sake of completeness
- Worse case: completeness issue not solved, irrelevant requirements with information harmful to system

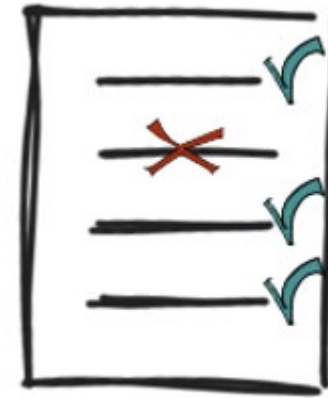
# Requirements Analysis: Requirements Prioritization



Limited Resources



=> Inability to satisfy all the requirements



⇒ Need to prioritize them

- Mandatory
- Nice to have
- Superfluous



# Example Quiz: Completeness and Pertinence

A software development team is creating a mobile app for online grocery shopping. They collected several requirements, including the ability to browse items, add items to a cart, make payments, and view order history. However, they also included a requirement to integrate a weather forecast feature that shows the weather in the user's location. Which issue does this extra feature represent?

- A) Lack of Completeness
- ☒ B) Lack of Pertinence
- C) Incomplete Requirements
- D) Redundant Requirements

# Requirements Specification: User and System Requirements

## User Requirements



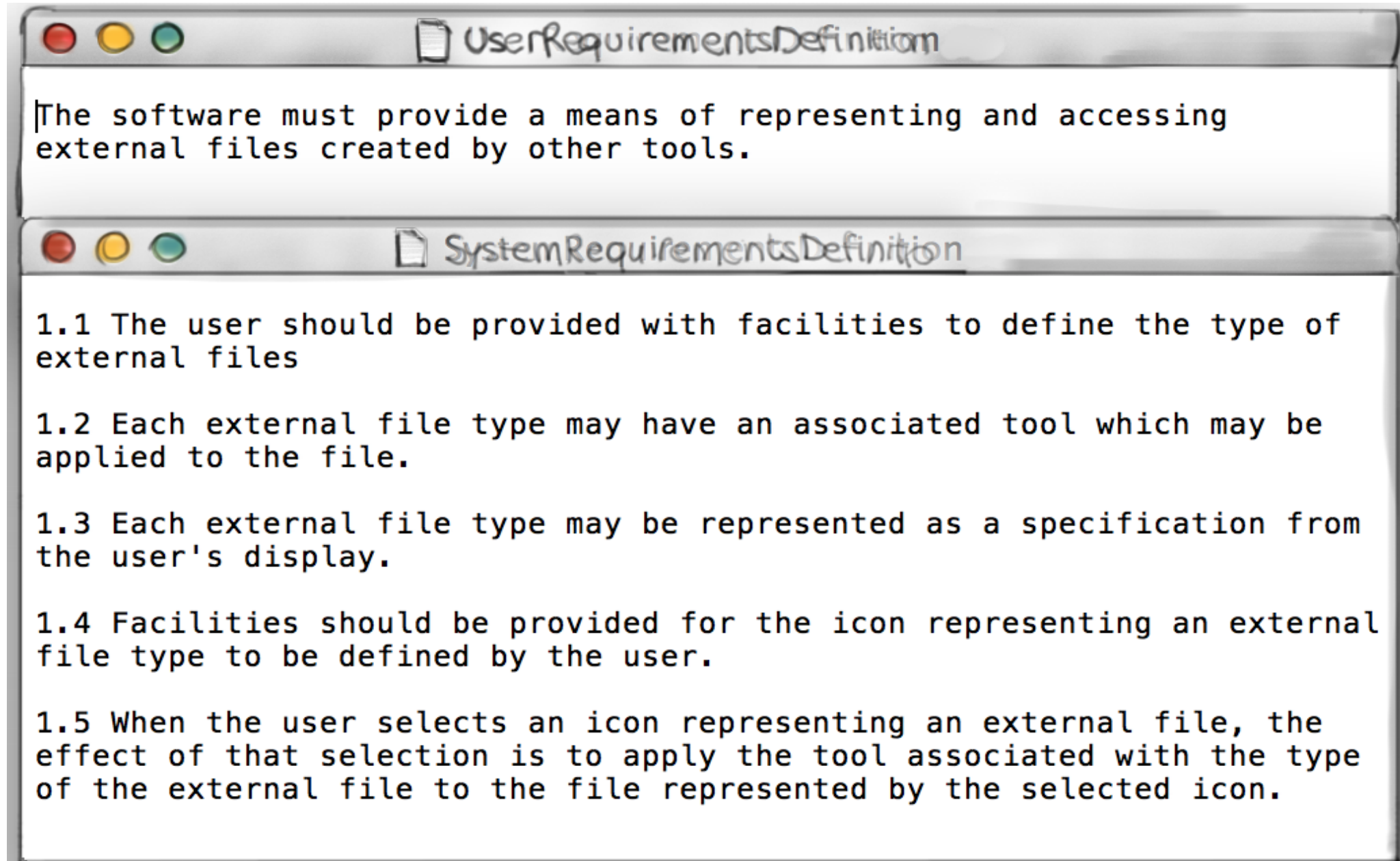
- Written for customers
- Often in natural language, no technical details

## System Requirements



- Written for developers
- Detailed functional and non-functional requirements
- Clearly and more rigorously specified

# User and System Requirements



# Requirements Specification: SRS

- The software requirements specification document is the official statement of what is required of the system developers.
- Should include both a definition of user requirements and a specification of the system requirements.
- It is NOT a design document. As far as possible, it should set of WHAT the system should do rather than HOW it should do it.
- Your project assignment uses the lightweight IEEE format of Software Requirements Specifications Document

# Example Quiz

A team is developing a new financial software tool for a bank. After conducting initial interviews, they find that different departments have conflicting requirements: one wants a simplified user interface for quick transactions, while another requires detailed analytics tools that clutter the interface. What should the team focus on next according to the principles of Requirements Engineering?

- A) Begin coding the software to meet the needs of one department.
- B) Develop a prototype incorporating all requested features.
-  C) Conduct further stakeholder analysis to prioritize and clarify requirements.
- D) Decide independently which department's needs are more critical.

# Bad → Good Requirement Activity

Is this a good requirement?

"The system should make it easy for students to find professors quickly."

Turn it into measurable and testable requirements?

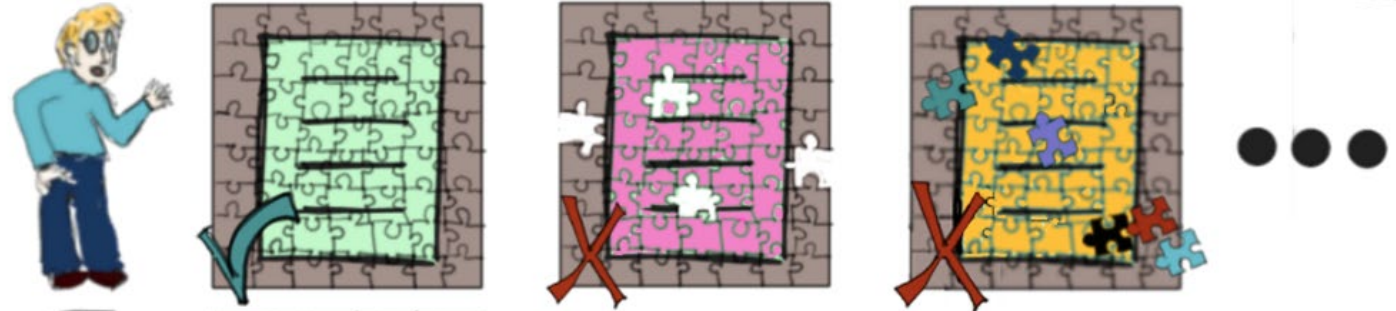
# Bad → Good Requirement Activity

- **FR-1:** The system shall allow users to search for professors by **name and department**.
- **FR-2:** The system shall display a message when **no professors match the search criteria**.
- **FR-3:** The system shall allow an administrator to **add, update, and remove professor information**.
- **NFR-1 — Performance:** The system shall display search results within **2 seconds for 95% of search requests**.
- **NFR-2 — Availability:** The system shall be available **99.5% of the time during normal university operating hours**.
- **NFR-3 — Capacity:** The system shall support at least **500 concurrent users** without exceeding a 3-second response time.
- **NFR-4 — Security:** The system shall require administrators to authenticate before modifying professor information.
- **NFR-5 — Usability:** At least 90% of first-time users shall successfully locate a specified professor within 60 seconds without assistance.
- **NFR-6 — Compatibility:** The application shall support the two most recent major versions of Chrome, Firefox, Safari, and Edge.

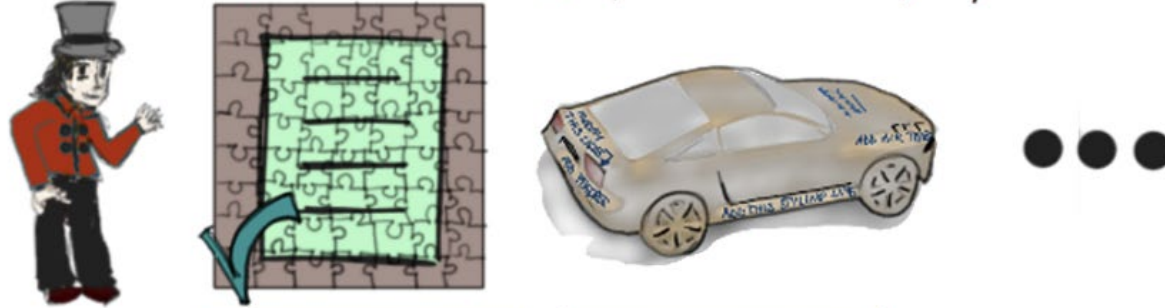
# Requirements Validation



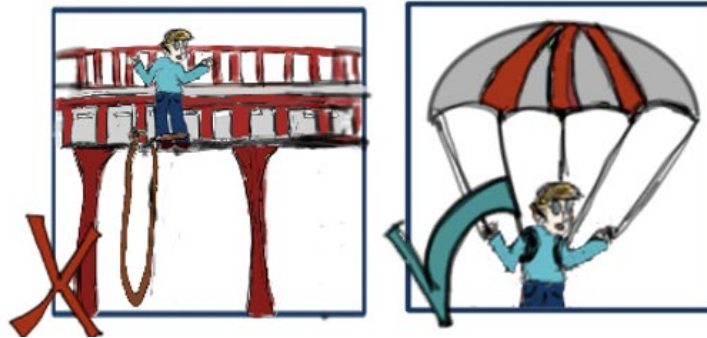
Verification



Validation



Risk Analysis



# Requirements Engineering Process

## 1. Requirements Elicitation

1. Consider functional and non-functional requirements

## 2. Requirements Analysis

1. User research
2. Pertinence and Completeness
3. Prioritize and Categorize

## 3. Requirements Specification

1. User and system requirements

## 4. Requirements Validation

1. Verification, Validation, Risk analysis

## 5. Management

# Requirements Engineering Process

